MILITARY SPECIFICATION

DIVERS CABLE ASSEMBLY COMMUNICATIONS AND STRENGTH

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of the communications and strength cable assembly for the MK 12 Mod O surface supported diving system (SSDS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-55442 - Cable Assemblies and Cord Assemblies, Packaging of.

STANDARDS

MILITARY

MIL-STD-147 - Palletized Unit Loads.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIP	592-4759110	- Connector Assembly, Pin.
NAVSHIP	592-4759112	- Cable, Comm/Strength.
NAVSHIP	592-4759153	- Cable Connectors Spares Kit.
NAVSHIP	592-4759159	- Comm. Cable/Strength Member Assy, 200 ft., W4.
NAVSHIP	592-4759160	- Comm. Cable/Strength Member Assy, 600 ft., W5.
NAVSHIP	592-4759161	- Connector Assembly, En.
NAVSHIP	592-4759162	- Comm. Cable/Strength Member Assy, 400 ft., W6.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Item composition. The communications and strength cable assembly shall be composed of a communication and strength cable in accordance with Drawing 592-4759112 which shall be terminated at one end by an Ell connector assembly in accordance with Drawing 592-4759161 and a pin connector assembly in accordance with Drawing 592-4759110 at the other end. The Ell and pin connectors shall be installed in accordance with the appendix to this specification, using special tools shown on Drawing 592-4759153.

3.3 Characteristics. The cable assembly shall be as specified in 3.3.1 through 3.3.1.2.

3.3.1 Performance. The performance of the communications and strength cable assembly shall be in accordance with all performance requirements specified in Drawings 592-4759159, 592-4759160 and 592-4759162 in addition to the requirements specified in 3.3.1.1 and 3.3.1.2.

3.3.1.1 Conductor direct current (de) resistance. Conductor dc resistance shall be less than 2 ohms for each 100 feet.

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3.3.1.2 Cable and connector pull strength. The cable and connectors, when assembled as a communications and strength cable assembly, shall withstand a minimum tension load of 1500 pounds for 1 minute, applied to each connector and the cable without electrical or structural degradation.

3.4 Construction. The cable assembly shall be constructed as specified in 3.4.1 through 3.4.2.

3.4.1 Fabrication and assembly. The communications and strength cable assembly shall be fabricated and assembled in accordance with Drawings 592-4759160, 592-4759162 and 592-4759159 as specified (see 6.2.1).

3.4.2 Standards of manufacture. Standards of manufacture shall be as specified on individual drawings.

3.5 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the inspections and tests specified in 4.5.2.1 through 4.5.2.4.

4.3.1 First article sample. The first article sample shall consist of one communications and strength cable assembly 200 feet in length, manufactured and packaged using methods, material and procedures identical to those proposed for production.

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in 4.5.2.2 and 4.5.2.3.

4.4.1 Quality conformance sample. Quality conformance samples shall consist of a random sample of three communication and strength cable assemblies selected as specified in 4.4.2. Nonconformance at completion of tests shall be cause for rejection. If two or more cables are rejected, an additional random sample shall be selected; if one or more of these samples are rejected, the entire lot shall be rejected.

4.4.2 Inspection lot. The cable assembly shall be assembled into identifiable inspection lots. Each inspection lot shall be manufactured on the same production lines by the same production techniques and to the same design with the material requirements within the same period not exceeding 6 weeks. Inspection lot identification shall be maintained from the time the lot is assembled through the time the lot is accepted and shall be traceable to the production lot from which the inspection lot was formed.

4.5 Methods of inspection. Inspections and test conditions shall be in accordance with 4.5.1 through 4.5.2.4.

4.5.1 Test conditions. Unless otherwise specified (see 6.2), all tests shall be conducted at the ambient conditions of the testing activity.

4.5.2 Inspections. Inspections shall be performed in accordance with 4.5.2.1 through 4.5.2.4.

4.5.2.1 Physical inspection. Parts, materials and assemblies shall be inspected to determine conformance to the drawings as specified in 3.2.

4.5.2.2 Electrical performance test. The communications and strength cable assembly shall be tested to verify that each individual conductor has continuity and is in conformance with 3.3.1.1 and all electrical requirements of the drawings as specified in 3.2.

4.5.2.3 Pull test and operating performance test. After being installed on the communications and strength cable assembly, the pin connector in accordance with Drawing 592-4759110 shall be subjected to a pull test. This pull test shall be done as follows: The cable shall be laid flat and secured at approximately 15 feet from the connector. The pin connector shall be threaded into a device that shall exert a tension load of 1500 pounds for a minimum of 1 minute. Each conductor’s dc resistance shall be measured from pin to pin during loading and shall not exceed 2.00 ohms for each 100 feet. The cable’s insulation resistance shall be measured during loading and shall be not less than 100 megohms for each 100 feet. The cable assembly shall be removed from the test fixture and visually inspected for damage. Any nonconformance shall be cause for rejection.

4.5.2.4 Leakage test. After completion of the tests specified in 4.5.2.1, 4.5.2.2 and 4.5.2.3, the cable assembly shall be submersed in salt water (representative of actual sea water) and connector cycled manually 20 times through 20 degree bends while submerged for a period of 6 hours. At the end of 6 hours, the cable shall be resubjected to the tests specified in 4.5.2.1 and 4.5.2.2 to verify that no leakage has occurred. Any nonconformance shall be cause for rejection.

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4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation-packaging and packing. Communications and strength cable assemblies shall be preserved-packaged level A or C and packed level A, B or C as specified (see 6.2) in accordance with MIL-C-55442.

5.2 Marking In addition to any special marking required (see 6.2), interior packages and exterior shipping containers shall be marked in accordance with MIL-C-55442 and the following:

"Communications and strength cable assembly
Part number: 80064-592-4759160 (Example)
Stock number: 8415-00-000-0000 (Example)
Serial number: (as assigned by contracting officer)
Manufacturer:
Contract number: N60921-00-0-0000 (Example)
Date of manufacturer: "

5.3 Palletization. When specified (see 6.2), the shipping containers packed as specified in 5.1 shall be palletized in accordance with load type 1 of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the appendix to MIL-STD-147. If the container or packages are of sizes which do not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

6. NOTES

6.1 Intended use. The communication and strength cable assembly is intended for use as a component of the MK 12 SSDS which will be used to support all surface-supported diving requirements of the U.S. Navy. It serves to conduct communication signals and provide the lifeline between surface and diver.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) If first article is required (see 3.1).
(c) Applicable fabrication Drawing (see 3.4.1).
(d) Test conditions, if other than specified (see 4.5.1).
(e) Levels of preservation-packaging and packing required (see 5.1).
(f) Special marking (see 5.2).
(g) Palletization, if required (see 5.3).

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6.3 First article inspection. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit as specified. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Communication strength cable assembly length not specified herein is also available on special order.

6.5 Subject term (key word) listing.

Cable
Communications
Direct current resistance
Diving system, surface supported

Preparing activity:
Navy - SH
(Project 4220-N300)

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APPENDIX

ASSEMBLY INSTRUCTIONS FOR THE MK 12
COMMUNICATIONS AND STRENGTH CABLE

10. SCOPE

10.1 Scope. This appendix provides the assembly instructions for the communications and strength cable assembly for the MK 12 Mod O surface supported system. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20• APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specification. The following specification forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL
QQ-S-571 - Solder, Tin Alloy; Tin-Lead Alloy; and Lead Alloy.

20.1.2 Government drawing The following Government drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSHIPS 592-4758613 - Lubricant.

(Copies of specifications and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

30. REQUIREMENTS

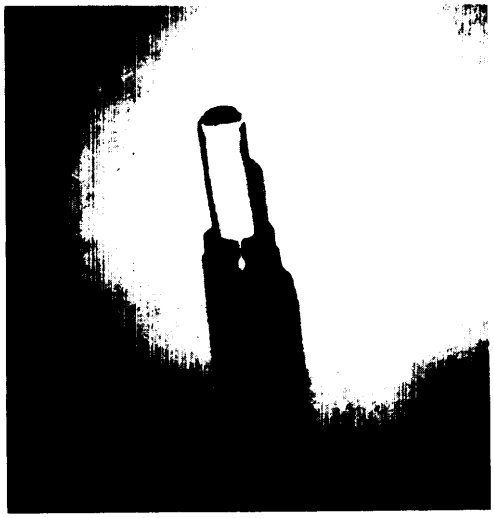
30.1 Assembly instructions. The communications and strength cable shall be assembled as specified in 30.2 through 30.5.

30.2 Steps A through M are identical for both diver’s end and topside end connectors.

30.3 Steps N through W apply to the divers end only.

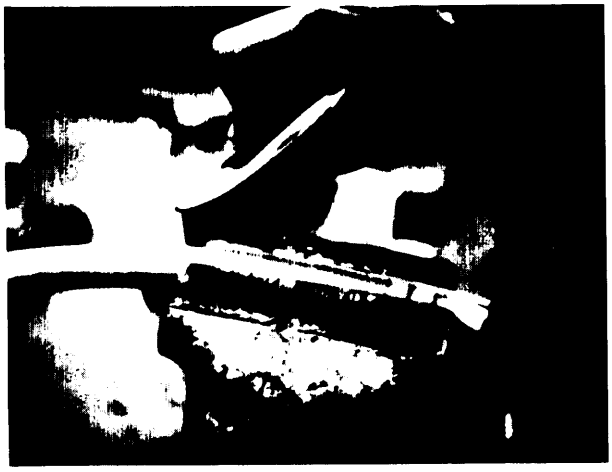
30.3.1 Step N1 applies to the topside end only.

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Picture 1

Step A. Slide boot assembly along cable to remove from work area (see picture 1). (Boot will be repositioned in step 1.)

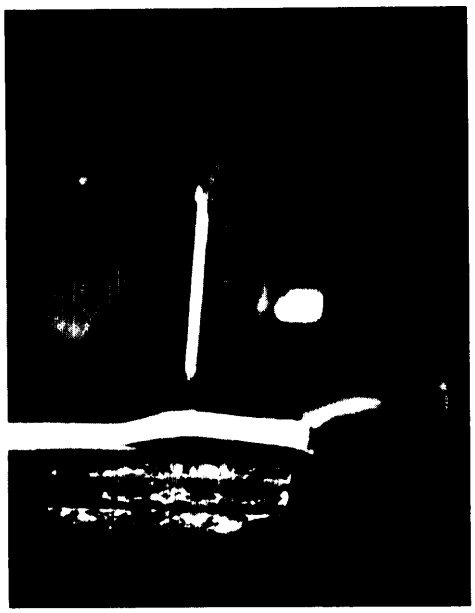


Picture 2

Step B. Place end of cable in vise and make four equally spaced 6-inch long cuts through outer polyurethane jacket to Kevlar braid (see picture 2). (Do not cut through Kevlar braid under outer cover.)

Step C. Turn cable so one strip is accessible above vise. Tighten vise. Strips on either side should be held by vise.

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Picture 3

Step D. Work a screw driver under tightly adhering jacket and pry it from Kevlar braid (see picture 3). Slide screw driver length of cut to separate jacket from Kevlar braid, peel jacket back (see picture 4). Lay jacket strip back into place and remove cable from vise.

Step E. Rotate cable to next strip, retighten vise, and repeat step D. Continue until all strips have been separated. Remove cable from vise. Cut off four outer jacket strips.



Picture 4

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Picture 5

Step F. Fold back Kevlar braid to expose inner polyurethane jacket and wire bundle. Place cable in cable chuck. Cut inner jacket through to wire bundle as close to Kevlar as possible (see picture 5). Slide jacket off cable and discard.



Picture 6

Step G. Pull Kevlar braid up around wire bundle, twist and tape end. Slide support tube over end of Kevlar and push down to outer polyurethane jacket (see picture 6).

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Picture 7



Picture 8

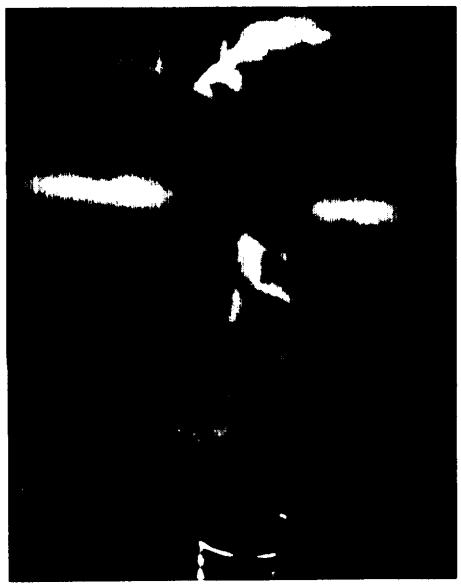
Step H. Place support tube installation tool over Kevlar braid. Hold Kevlar braid through opening in support tube installation tool with pliers to prevent bunching. Using a hammer on the top of the support tube installation tool, drive the support tube down 1-1/4 inches between outer jacket and Kevlar braid (see picture 7). Leave about 1/4 inch of support tube exposed (see picture 8). Pull installation tool off. Trim outer jacket evenly.

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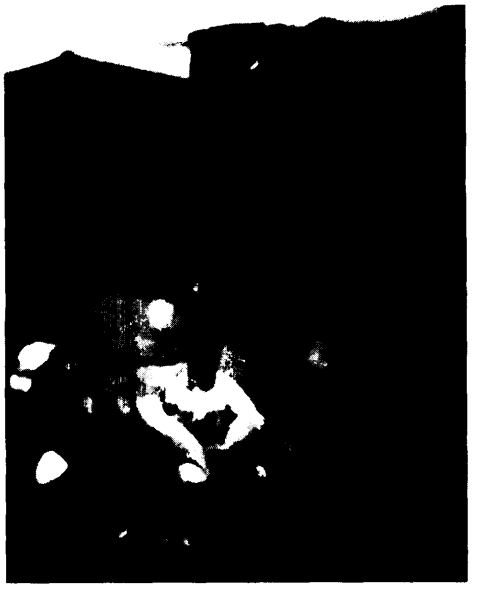
Picture 9

Step I. Slide boot assembly as far up support tube as possible. Place boot pulling tool over Kevlar braid and thread female end onto boot assembly. Tighten boot assembly by turning male end of pulling tool while holding female end in vise (see picture 9). Tighten until boot is properly seated against support tube. Remove pulling tool (see picture 10).



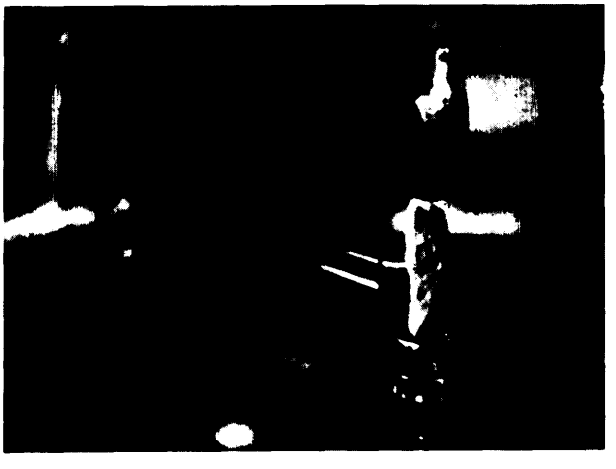
Picture 10

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Picture 11

Step J. Remove tape and fold Kevlar braid down over support tube. Slide wedge over wire bundle and push down between Kevlar and wire. Place wedge installation tool over wire bundle. Holding Kevlar braid back tightly, hammer wedge installation tool. This will push wedge between Kevlar and wire bundle into support tube (see picture 11). Remove wedge installation tool. Pull Kevlar up and tape.



Picture 12

Step K. Slide mechanical termination nut over cable end and thread into support tube. Using a torque wrench, tighten nut to 50 inch-pounds (in-lb) (see picture 12).

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Picture 13

Step L. Untape Kevlar braid and cut braid off close to nut (see picture 13).



Picture 14

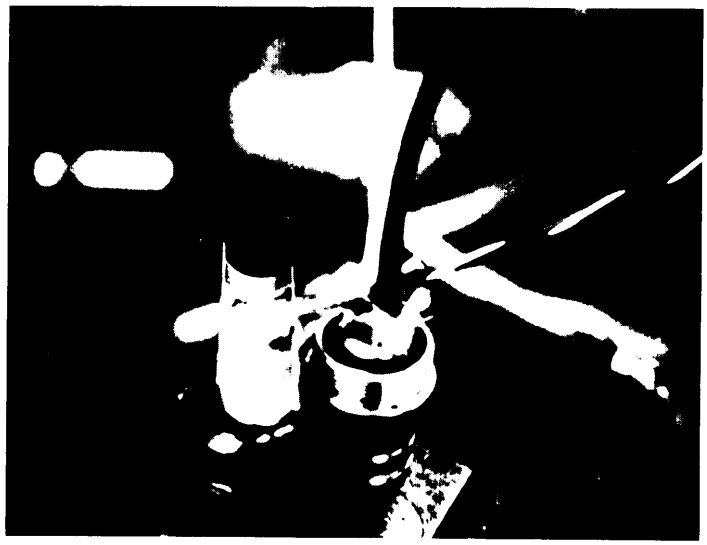
Step M. Remove cellophane cable filler and separate wires (see picture 14) as follows: Two twisted shielded pairs of polyester (PE) wire (each pair consists of two wires inside, one black and one white insulated copper wire; one pair is insulated on the outside with black PE and the other is insulated with white PE; both are armor shielded in woven wire); two twisted sets of three wires (each set contains one black and one white insulated wire with the other wire being green in one set and red in the other); and four single wires (one each: black, white, green and red).

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Picture 15

Step N. Check red-white-black triple for continuity. If these wires do not check out, check green-white-black. cut off twisted triple which you will not be using and the four single wires. Unbraid shielding from both shielded wires (see picture 15), twist together and solder to one of the spacer tubes as shown on picture 16.



Picture 16

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Picture 17

Step O. Slide pin connector boot over cable wires as shown on picture 17.



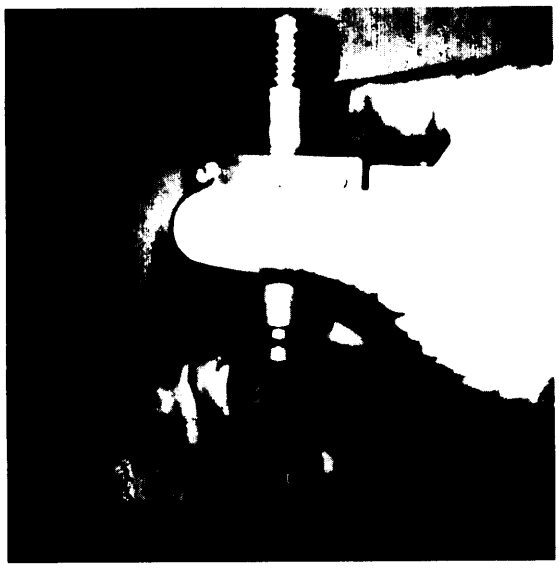
Picture 18

Step P. Solder wires to seven-pin connector as shown on picture 18 and figure 2.

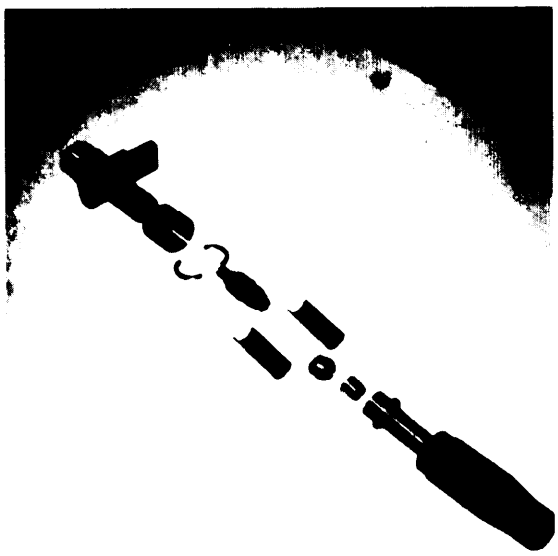
Step Q. Install new O-ring in groove of pin connector.

Step R. When pins are cool, lubricate with Halocarbon lubricant, in accordance with Drawing 592-4758613. Slide boot up over soldered pins. Seat boot on connector bottom. Apply a light film of lubricant to outside of pin connector.

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Picture 19



Picture 20

Step S. Slide connector nut onto shell and hold in place. Install new O-ring onto bottom of shell.

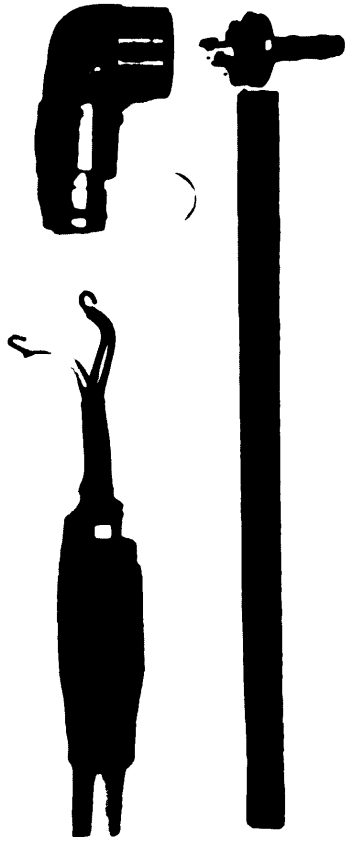
Step T. Place tube spacers on each side of cable with bottoms fitting into outer rim of support tube and tops fitting under edge of pin connector boot.

Step U. Align key (in pin connector) and keyway (in shell). Ensure that roll pin in base of shell aligns with one of support tube holes.

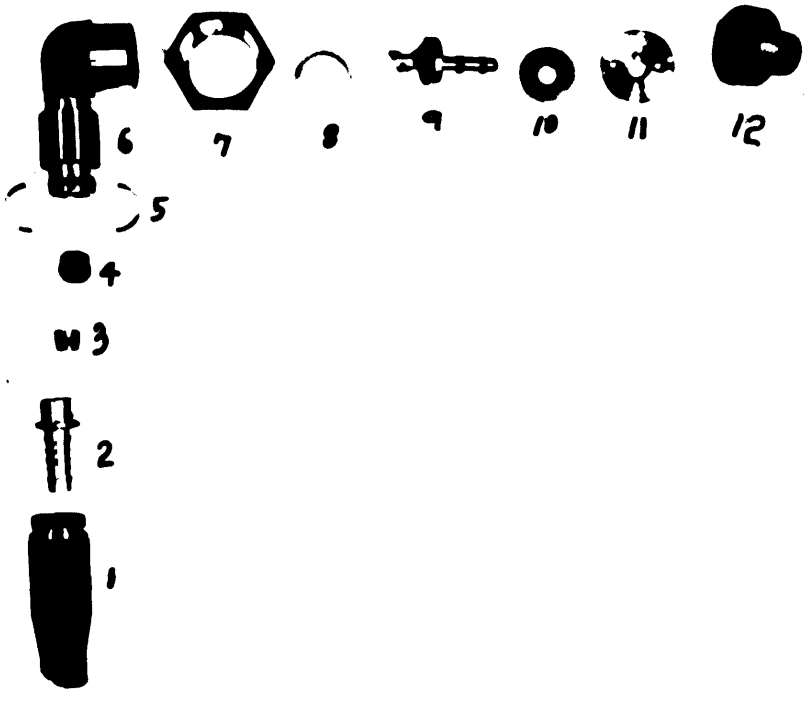
Step V. While holding tube spacers around wiring, slide shell over connector boot assembly until pin fits into receptacle hole in support tube (see picture 19). Ensure that shell and boot assembly are secured such that they do not turn.

Step W. Install both retainer ring halves into shell and slide connector nut down over them securing them into place. Hand tighten connector nut. Lock into place and tighten setscrew. Remove from cable chuck. See picture 20 for proper parts setup.

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Picture 21



Picture 22

Step N1. Locate the two shielded pairs (one with white jacket, one with black). Strip off shielding and jacketing exposing twisted pairs. Strip off sufficient insulation and solder wires together, white to white and black to black using S/N solder in accordance with QQ-S-571. Unused conductors should be trimmed to the outer end of the mechanical termination nut as shown on picture 21. Pass the leads through items 6, 7 and 8 as shown on picture 22, then connect the white leads to the longer terminal and the black leads to the shorter terminal of the telephone plug (see picture 22 and figure 3). Assemble the remaining parts as shown on picture 22. Note that in fitting the slide nut and shell (item 6) into the support tube (item 2), the guide pin must be mated into the receptacle hole in the support tube. The mechanical termination nut shall be torqued to a value of 45 to 55 in-lbs, all other torques shall be as required for proper assembly. Continuity checks shall be made between the white leads and pins 2 and 6 on the diver's end and between the black leads and pins 1 and 5 on the diver's end.

1. Boot
2. Support tube
3. Wedge
4. Mechanical termination nut
5. Retaining ring
6. Slide nut and shell
7. Nut
8. Leather ring
9. Telephone plug
10. Leather washer
11. Telephone plug retainer
12. Cap

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30.4 Special tools. The special tools required to work on the cable shall be as shown on figure 1.

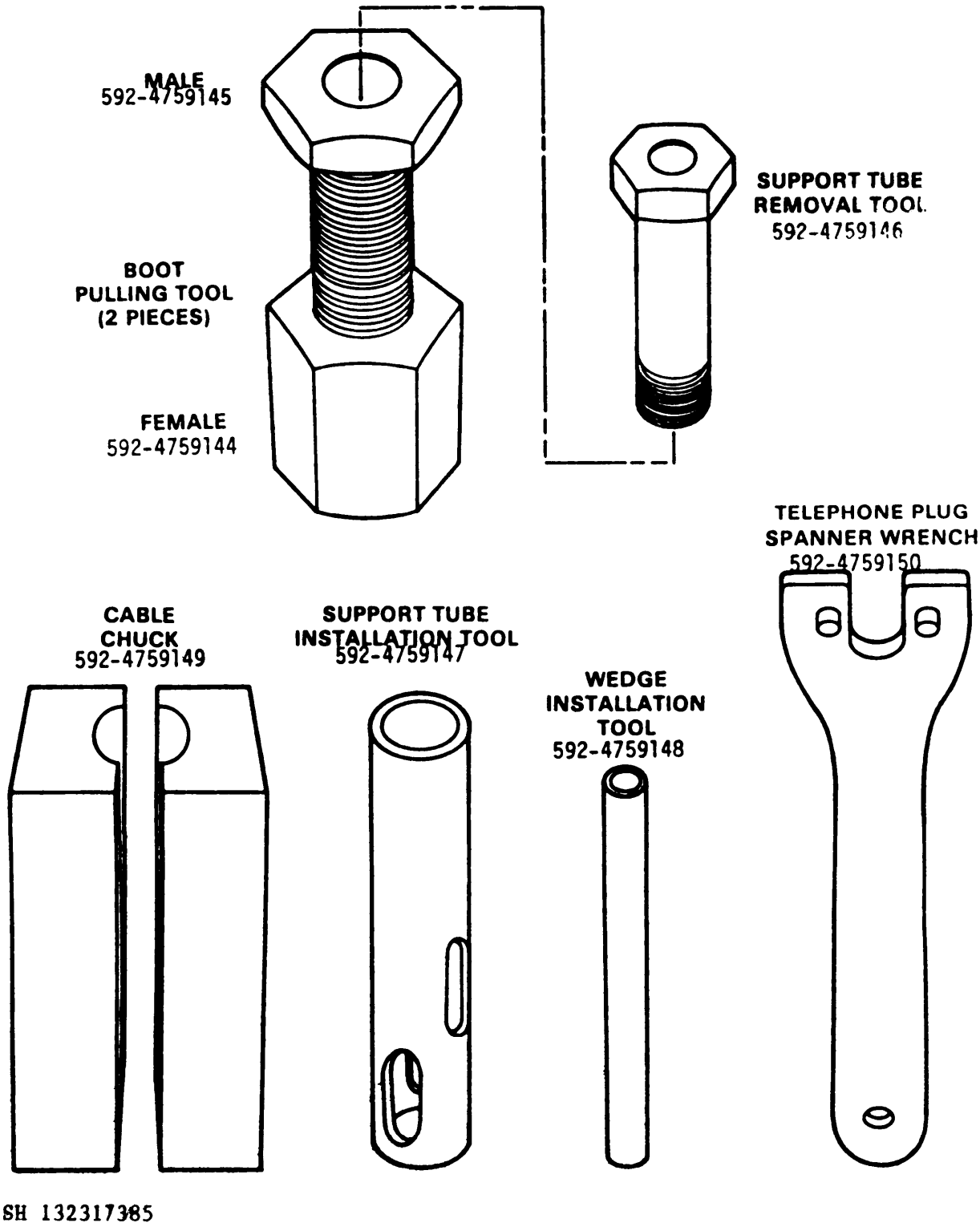


FIGURE 1. Special tools.

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30.5 Hook-up The hook-up for the diver's end and the topside end shall be as shown on figures 2 and 3.

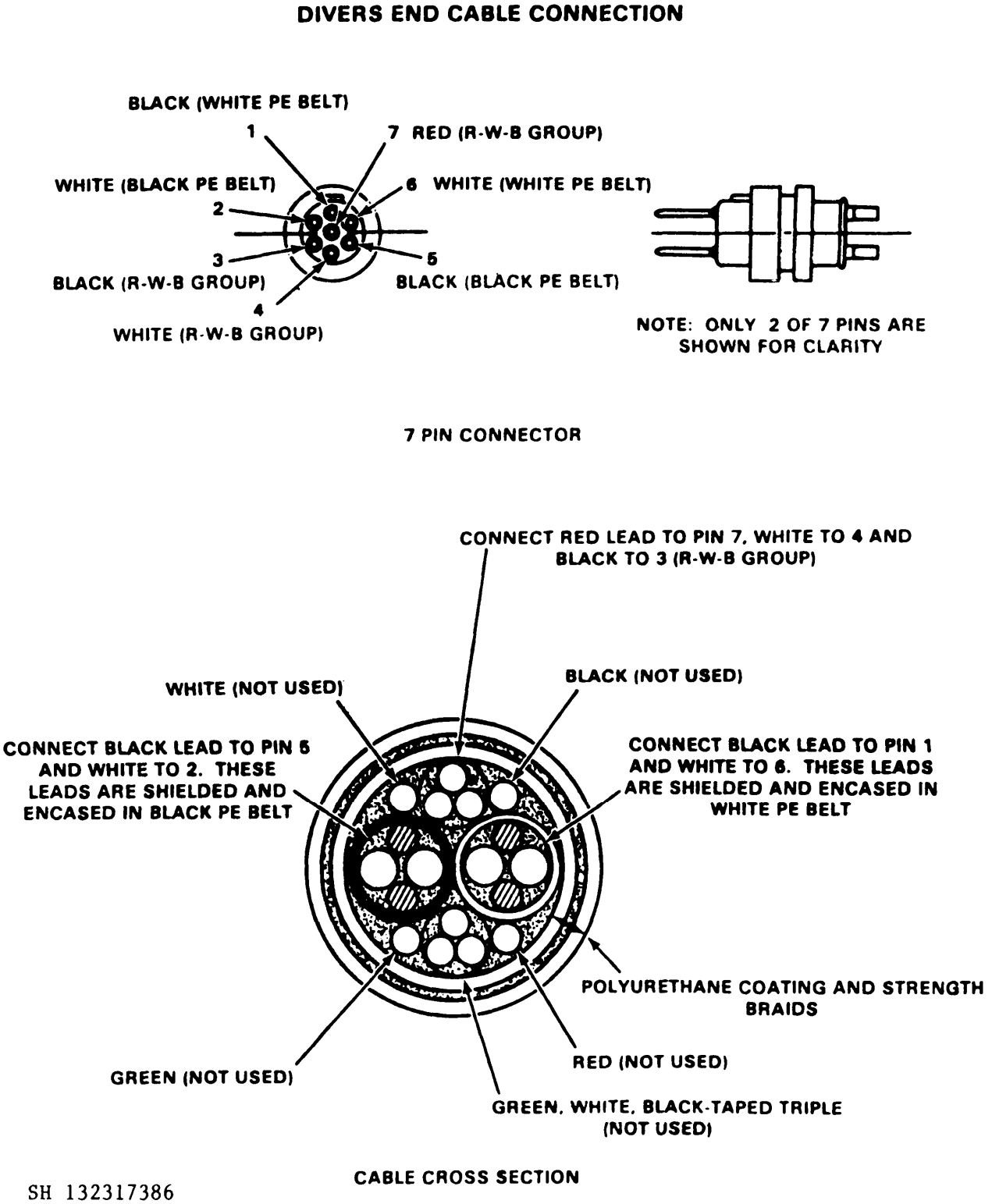
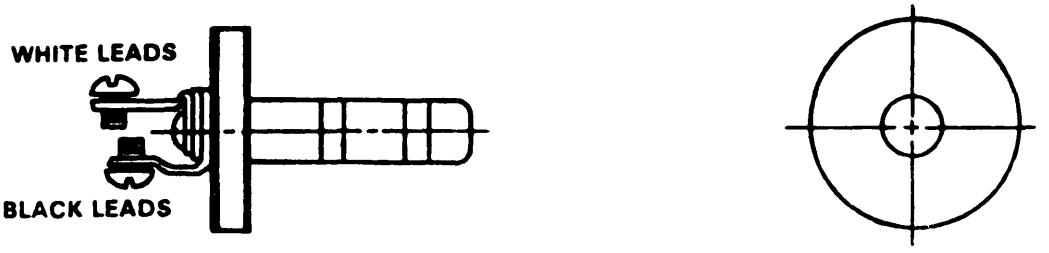


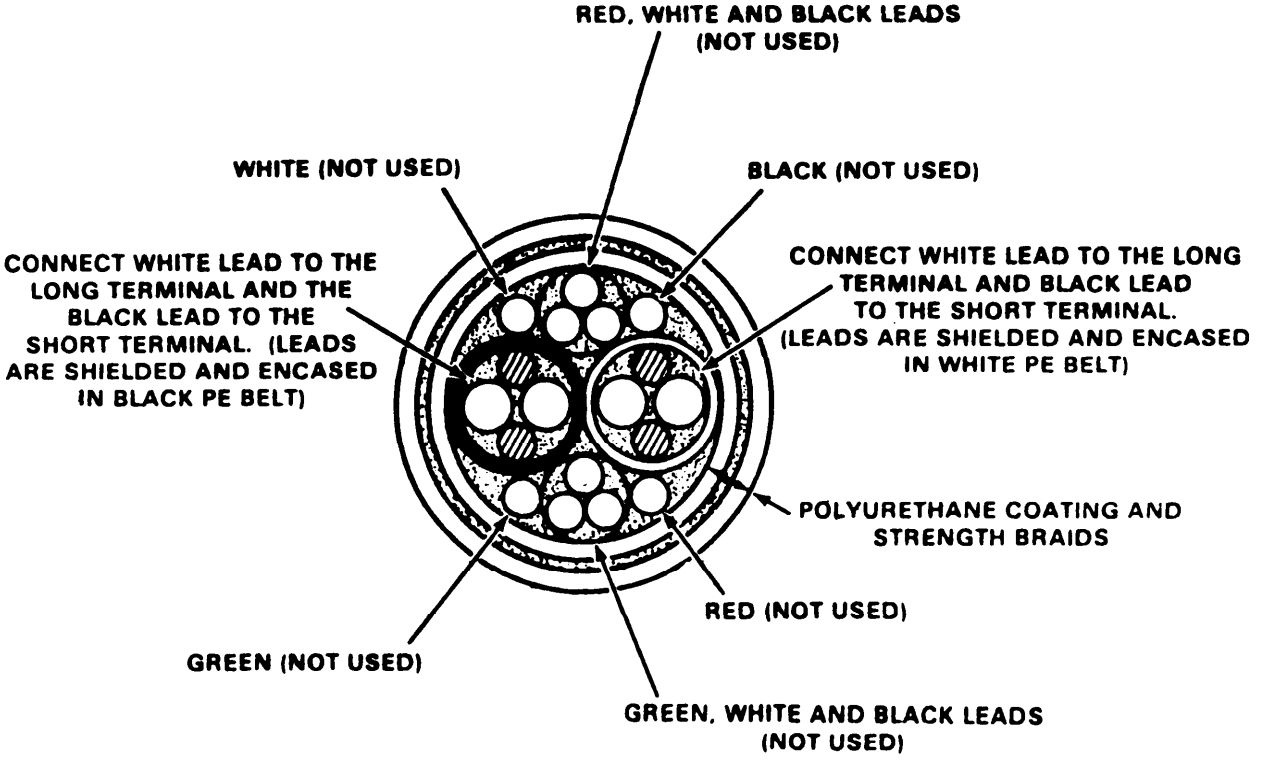
FIGURE 2. Hook-up for diver's end.

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TOPSIDE CABLE CONNECTION



TELEPHONE PLUG



CABLE CROSS SECTION

SH 132317387

FIGURE 3. Hook-up for topside end.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-D-24684(SH)	DIVERS CABLE ASSEMBLY COMMUNICATIONS AND STRENGTH

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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